

Correlation-Aware Probe Planning

for Cloud Identity Attack Paths

beliefs

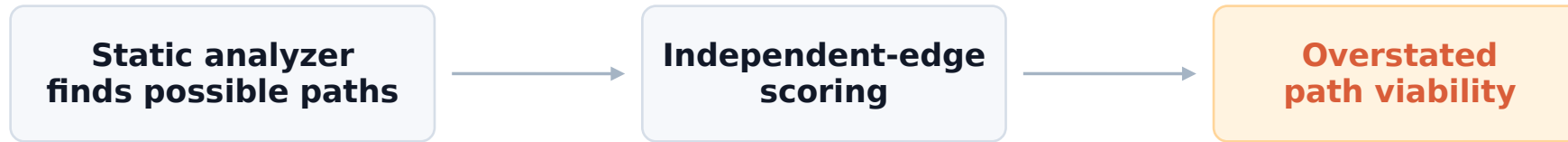
constraints

ELG probes

**Shared controls
create shared-
fate edges.**

Static reachability misses shared failure modes

A path graph can look independent even when governance is shared.

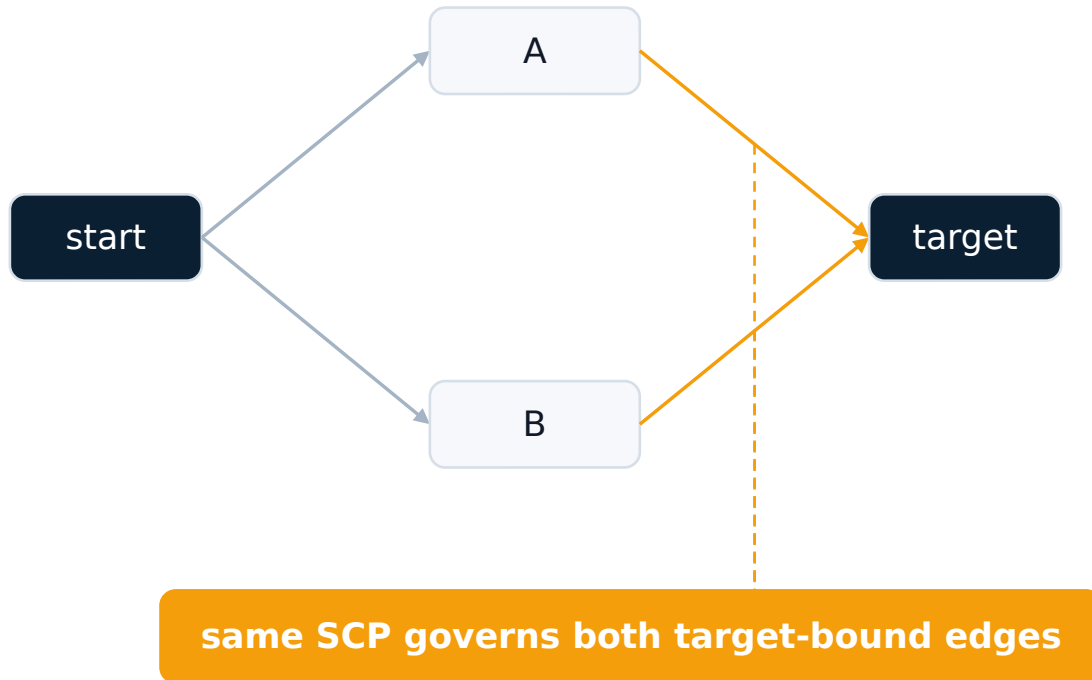


One SCP or trust condition can govern many edges at once.

A probe on one edge can update the belief state for a whole component.
The planning problem is objective-level uncertainty, not just local edge uncertainty.

Small diamond, large probability shift

Correlation changes the estimate without changing the static graph.



Naive independent

55%

Correlation-aware

1.2%

**≈46×
reduction**

Mechanism: a strong-looking edge shares a constraint group with a blocked representative.

ARF-RT: from static graph to next best probe

Deterministic replay over beliefs, constraints, correlation, paths, and EIG.

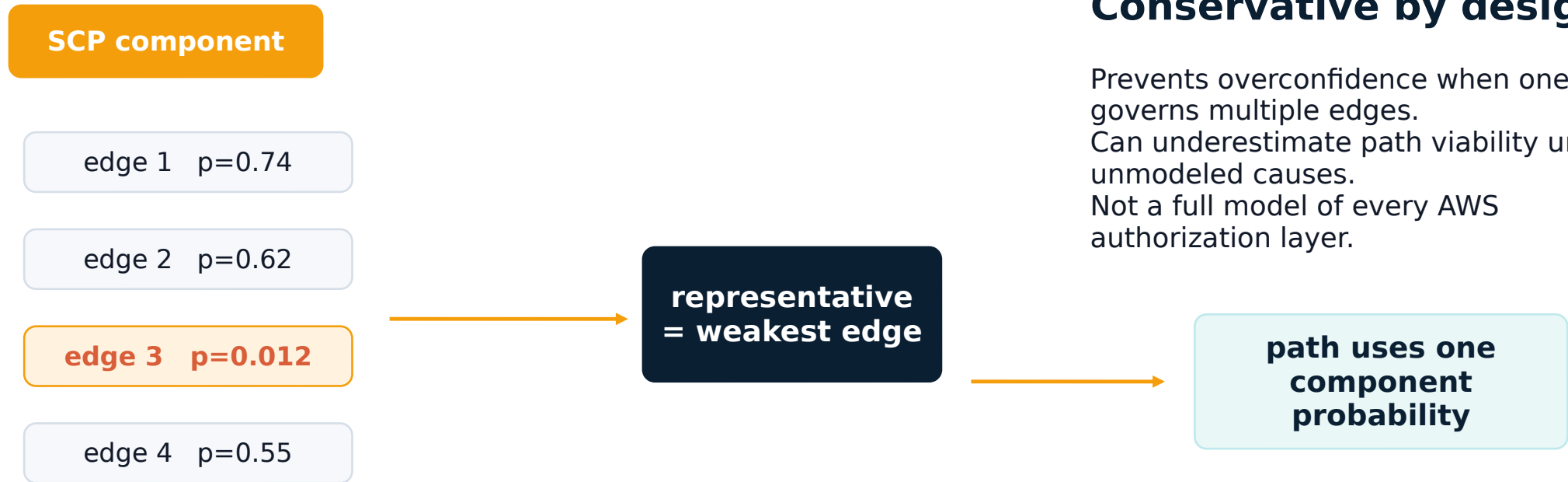


Same input → same probabilities → same run hash

Integer updates and Q8 fixed-point arithmetic avoid floating-point drift.
Constraints are explicit graph objects.
Each probe fork reruns the full pipeline before scoring entropy change.

Worst-case representative substitution

For each correlated component, score the path through the weakest representative.



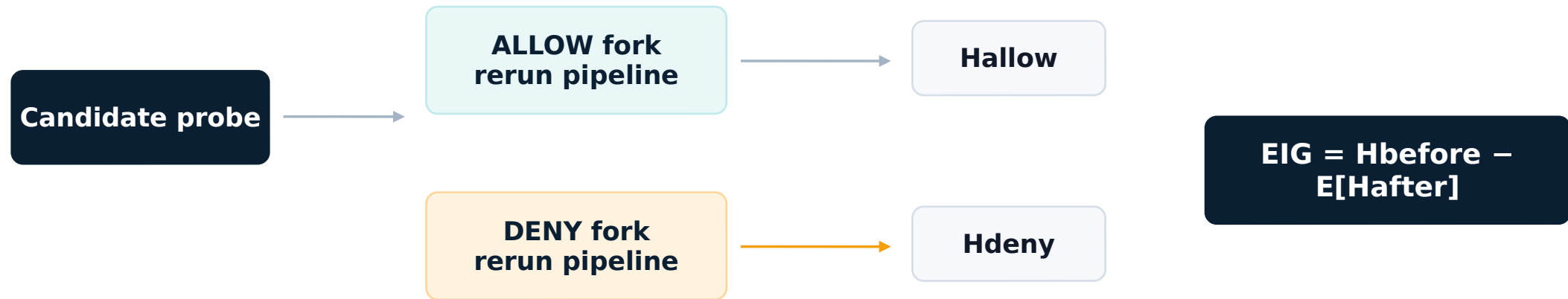
Conservative by design

Prevents overconfidence when one policy governs multiple edges.
Can underestimate path viability under unmodeled causes.
Not a full model of every AWS authorization layer.

Safe claim: conservative within the modeled shared-fate constraint classes. (Edge values illustrative; see slide 3 for the verified diamond fixture.)

EIG selects the probe that changes objective uncertainty

The planner simulates both outcomes, then chooses expected entropy reduction.



Not “most uncertain edge.” Not “most central edge.” Objective-level entropy.

Primary replay fixture: EIG outperforms local heuristics

EIG removes 0.2361 of 0.3013 nats — ~78% of baseline path uncertainty — in 5 probes. (Multipliers below are one instance; the absolute reduction is the robust headline.)

Policy	CumRIG@1	@2	@3	@5
EIG	0.1336	0.1827	0.2087	0.2361
Random	-0.0004	0.0128	0.0123	0.0119
Uncertainty	0.0134	0.0134	0.0134	0.0134
Centrality	-0.0020	-0.0025	-0.0029	-0.0033

19.8×
vs random

17.6×
vs
uncertainty

Centrality goes negative
when structural prominence $\neq$
residual uncertainty

Fixture: 14 principals · 16 edges · 2 SCPs · 2 trust conditions · 11 observations

SVII-E control: with constraints removed and uniform priors, EIG/uncertainty drops to $\approx 1.1\times$ — advantage attributable to correlation modeling.

Complex fixture: one SCP component remains highest-value probe

A richer constructed environment makes the leverage operationally visible.

4 business areas, 8 AWS accounts
24 principals, 33 edges, 6 constraints
15 paths across 4 high-value targets

**Top probe resolves
10 SCP-governed
edges**

Component	Type	Edges	Probe value
BlockCrossAccountAssume Prod	SCP	10	EIG $\approx$ 1.072
Other trust / singletons	Mixed	1 ea.	Much lower

One SCP probe, full Top-K:
2.70 $\rightarrow$ 1.09 nats $\approx$ 60%

Safe claim: the planner prioritizes a shared governance bottleneck; magnitude is corrected in v1.2.

Cross-environment trust weakening

Synthetic, prior-only OIDC/GitHub enrichment changes trust-edge priors.



Gate-bypass finding weakens the trust prior

Path	Before	After
infra-terraform	0.25	0.375
payment-api	0.167	0.333
db-migrations	0.167	0.250

**3 compromised gates
3 priors adjusted
0 direct observations**

Composition demo, not field validation.

Takeaway: attack paths are not always independent-edge graphs

ARF-RT sits after static analysis and asks what to test next.



What this demonstrates

Shared governance constraints change path probabilities.
Correlation-aware EIG can prioritize probes with multi-path leverage.
The evaluation is controlled and replay-based, not field validation.

Core claim: model shared controls before deciding what to test next.

Thank you

Artifact (v1.2): zenodo.org/records/20186190